

Meta Enhanced Brain Tumor Segmentation with Partial MRI Modalities Through Adaptive Fusion and Bi-Level Optimization

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Abstract

Brain tumor segmentation is fundamental to clinical diagnosis; however, the frequent unavailability of certain Magnetic Resonance Imaging (MRI) modalities during training or inference poses a significant challenge. Existing studies typically employ all modalities, a practice often impractical in real-world scenarios due to acquisition errors, artifacts, time constraints, or protocol variations. To overcome this limitation, we propose METAL, a bi-level optimized meta-learning framework for brain tumor segmentation that employs the nnUNet backbone, attention-based fusion, and an adversarial modality discriminator to handle partial MRI modalities. The framework incorporates lightweight adapters for fast, sample-specific adjustments and adopts a bi-level training strategy to optimize the network with both partial- and full-modality inputs. We evaluated our model on three datasets, including Brain Tumor Segmentation (BraTS) 2019 (high-grade gliomas and lower-grade gliomas), BraTS2020, and BraTS2021, achieving high DSC Similarity Coefficient (DSC) of 92.31%, 90.53%, and 88.37%; 90.62%, 88.71%, and 85.92%; 92.59%, 90.86%, and 87.12%; and 93.12%, 91.20%, and 88.29% for Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET), respectively. Our method outperforms state-of-the-art transfer learning models by 5–10% and exceeds previously reported approaches in the current literature. These experimental results demonstrate the effectiveness and robustness of our proposed approach, confirming its reliability in practical applications.

Keywords: Tumor Segmentation; Partial Modalities; Bi-level Optimization; Meta-learning; Attention-based Fusion; Adversarial Training.

1 Introduction

A brain tumor represents one of the most life-threatening medical conditions, affecting people of all age groups and sexes [1, 2]. The incidence of brain tumors in children is reported to be on the rise, possibly attributable to the increased use of digital technologies [1, 3]. Precise segmentation of brain tumors is crucial both in neuroscience research and clinical practice, facilitating applications such as population-level analyses of cortical structures and personalized diagnosis of neurological diseases [4, 5]. However, brain MRI presents considerable challenges, as the structural and functional characteristics of the brain undergo spatio-temporal dynamic changes throughout the human lifespan, leading to longitudinally heterogeneous appearances and varying levels of segmentation difficulty [4, 6, 7]. In clinical settings, reliable brain tumor segmentation is usually based on complementary data from

various MRI modalities, including FLAIR, T1ce, T1, and T2 [8, 9, 10]. However, the common lack of certain modalities, especially T1ce due to contrast agent allergies, poses practical difficulties [8, 11]. In addition, in some clinical settings, only a single modality, such as T1, T2, or FLAIR, may be available for diagnostic evaluation [12, 8]. This highlights the need to develop strong, modality-independent techniques that can achieve high-level performance even when some modalities are absent, either during training or inference [13, 14, 15].

Various image processing and machine learning techniques have been suggested to enhance brain tumor segmentation [16, 17, 4, 8, 18, 19, 20]. In the context of medical image segmentation, meta-learning frameworks and methods addressing missing input modalities have attracted considerable attention, with models such as Self-Sampling Meta SAM (SSM-SAM) [16], MetaSwin [17], dual meta-learning (DuMeta) [4], and Learnable Sorting State Space Model (LS3M) [8]. These

models are designed to capture key structural and semantic information across different imaging phases. In addition, several methodologies have been developed specifically for brain tumor segmentation. One category comprises hybrid and encoder-decoder architectures, including Javeria Amin Segmentation Method (JASM) [18], Multiscale Segmentation Network (MS-SegNet) [19], and region-of-interest detection algorithm (ROI-DA) [20], which focus on multiscale feature extraction and region-specific representations. Another category consists of attention-driven networks with integrated fusion modules, such as multi-perspective extraction and dense attention network (MPEDA-Net) [21], brain tumor segmentation network (BSAU-Net) [22], guided attention inception residual U-Net (GAIR- U-Net) [23], and modified bi-directional ConvLSTM U-Net (M-BDCU-Net) [24], which facilitate the integration of diverse information from multiple images.

Recent developments in brain tumor segmentation have made progress; however, several challenges remain unaddressed. Many existing models [17, 18, 21, 19, 20, 22, 23, 24] do not explicitly enforce the learning of modality-independent representations, which restricts their ability to focus on tumor-relevant features rather than modality-specific patterns. This limitation reduces robustness when certain MRI sequences are missing, a situation that frequently occurs in real-world clinical settings. In addition, current approaches generally lack mechanisms to dynamically adjust the network based on available input features, making them less adaptable to incomplete or heterogeneous data. Their training strategies also tend to assume consistent input modalities and do not consider varying combinations of available sequences, which can hinder generalization across diverse patient populations and imaging conditions. Collectively, these limitations highlight the need for models that can learn flexible, modality-independent representations, adapt efficiently to available inputs, and maintain robust performance even under incomplete imaging scenarios.

This motivated the development of our proposed model, METAL, a novel segmentation framework capable of accurately segmenting brain tumors even in the absence of one or more modalities. The architecture is built upon the nnUNet backbone and integrates a bi-level meta-learning strategy with attention-based fusion to enhance tumor segmentation. Specifically, the framework employs a bi-level training strategy where the inner loop adapts lightweight modality-specific adapters for handling partial-modality inputs, while the outer loop updates the global network using complete multi-modality scans. To further improve robustness, an adversarial modality discriminator is incorporated, encouraging the encoder to learn modality-independent latent representations. The inclusion of lightweight adapters enables rapid adaptation to the available input features, ensuring consistent performance under missing-modality conditions. The major contributions of our study are given below:

- Introduces a bi-level meta-learning framework that integrates the nnUNet backbone, attention-based fusion, and adversarial modality discriminator, and a lightweight adapter to effectively handle missing MRI modalities in brain tumor segmentation. Our proposed model achieves accurate segmentation even when some MRI modalities are missing, addressing a common challenge in clinical practice.
- Incorporates an adversarial modality discriminator to encourage the network to learn modality-independent latent representations; by encouraging the encoder to focus on tumor-relevant features while minimizing modality-specific patterns, this method improves robustness even when several modalities are absent.
- Integrates lightweight adapters that adjust the network based on the available input features, enabling rapid, sample-specific adaptation without retraining the entire network. This design ensures computational efficiency while maintaining high segmentation accuracy under incomplete modality conditions.
- Employs a bi-level training method, where the inner loop adjusts the adapters for partial-modality inputs, and the outer loop updates the global network using full-modality scans, to ensure generalization across varying combinations of available MRI modalities.
- We evaluated the proposed framework on three benchmark datasets, including BraTS2019, BraTS2020, and BraTS2021. Experimental results demonstrate that our approach achieves superior segmentation performance and remains effective in scenarios with missing modalities in brain tumor segmentation.

The rest of this paper is organized as follows. Section 2 reviews existing works on meta-learning and brain tumor segmentation. Section 3 details the methodology of our proposed model, including problem formulation and overall model architecture. Section 4 presents the experimental results, including performance assessment, cross-dataset evaluation, and performance comparison. Section 5 discusses the implications of the findings and potential future directions. Finally, Section 6 summarizes our key contributions.

2 Related Works

In this section, we present a review of recent studies in meta-learning, missing-modality handling, and brain tumor segmentation.

2.1 Meta-learning and missing modality segmentation

Recent works [13, 16, 17, 4, 25, 8] have explored meta-learning approaches and methods for handling missing modalities to improve segmentation accuracy. Initially, Konwer et al. [13] developed a meta-learning architecture (MLA) that addresses missing modalities by generating particular encoders for each modality and distributing parameters. It incorporated a feature aggregation module that combines available and missing modality features into a single representation, which is then processed by a Swin-UNETR decoder. It achieved DSC of 88.74, 81.63%, and 65.27% for the WT, TC, and ET regions, respectively. In another study, Leng et al. [16] proposed the SSM-SAM, which incorporates a meta-optimized fast gradient descent optimizer. It also included a self-sampling module to create properly aligned visual prompts and an attention-based decoder to achieve better segmentation. However, the system focuses on few-shot adaptation and rapid alignment while lacking explicit methods for modality-invariant feature learning. This model obtained DSCs of 73%, 53.08%, and 41.69% for WT, TC, and ET, respectively.

Similarly, Lee et al. [17] presented the MetaSwin model, which improves SwinUNETR by replacing attention-based components with spatial pooling methods inspired by the MetaFormer concept. This model also included Squeeze-and-Excitation blocks following every MetaSwin block in the encoder and decoder to enhance segmentation performance. However, it relies heavily on attention mechanisms, making it less efficient and less adaptable to varying input modalities or feature distributions. In contrast, Sun et al. [4] presented the DuMeta framework, designed to capture longitudinally consistent representations that remain stable during fine-tuning. This method incorporates a well-initialized task head optimized using meta-initialization, along with a plug-and-play feature extractor to obtain anatomical representations through meta-feature learning. To improve longitudinal consistency, two class-aware regularization techniques are also included. However, while the framework emphasizes longitudinal consistency, it does not explicitly address variations in available modalities.

In another paper, Fang et al. [25] presented the Hierarchical In-Out Fusion network (HIOFN), which handles incomplete multimodal inputs using an out-fusion block with self-attention at shallow layers and dual-domain fusion at deep layers. They also incorporated an in-fusion block with stacked Transformer-CNN blocks to capture modality-specific features and sequential intra- and inter-modal fusion to generate a shared representation for segmentation, achieving DSC of 84.97%, 74.42%, and 63.83% for WT, TC, and ET, respectively. Finally, Zhang et al. [8] introduced the LS3M, which reorders input sequences using differentiable permutation matrices to preserve modality-specific information and spatial relationships. It incorporated the Series State Space Model block to capture both short and long-range dependencies to

represent intra- and inter-modal interactions accurately. It obtained DSC values of 88.22%, 79.76%, and 63.6% for WT, TC, and ET, respectively.

2.2 Brain tumor segmentation

Recent studies [18, 21, 19, 20, 22, 23, 24] have focused on brain tumor segmentation, focusing on improving the precision of tumor region identification. To begin with, Amin et al. [18] proposed the JASM, a dual-model approach combining U-Net and a Region Proposal Network built on ResNet-50 for brain tumor segmentation in MRI. It effectively segmented core and edema regions, achieving an average DSC of 91% on both BraTS2020 and BraTS2021 datasets. In the other paper, Luo et al. [21] proposed MPEDA-Net, designed to allocate appropriate attention to each point while effectively combining feature information. It integrated three distinct convolutional kernels along with specialized operations to enhance feature representation. It achieved DSC of 82.44%, 91.38%, and 88.27% for ET, WT, and TC on BraTS2018, and 83.73%, 91.87%, and 88.71% on BraTS2019, respectively.

Similarly, Sachdeva et al. [19] developed MS-SegNet to segment heterogeneous brain tumors, specifically gliomas, from multimodal MR images. It incorporated a multi-scale feature extraction block to capture fine-grained information, such as tumor boundaries and the edges of necrotic regions. Evaluation results indicated that MS-SegNet achieved DSC of 81%, 91%, and 83% on BraTS2020, and 86%, 92%, and 84% on BraTS2021, for ET, WT, and TC, respectively. While Schwehr et al. [20] implemented an ROI-DA during MRI pre-processing to emphasize relevant features and remove redundant data. Subsequently, a fully convolutional autoencoder with channel-wise attention and attention gates performed segmentation, followed by test-time augmentations and an energy-based model for voxel-level uncertainty estimation. It was evaluated on BraTS2019, BraTS2020, and BraTS2021 data sets, achieving a DSC of 84.55%, 88.52%, and 90.82%, respectively. In contrast, Cao et al. [22] introduced BSAU-Net, which combines edge and spatial attention mechanisms. It used an edge feature extraction module based on the Sobel operator to preserve tumor boundaries and a spatial attention module to capture global feature relationships. To address class imbalance, the network employed a specialized loss function, BADLoss. On the BraTS2018 and BraTS2021 datasets, it achieved DSC of 75% and 76%, respectively.

In another study, Rutoh et al. [23] proposed GAIR-U-Net for multimodal MRI brain tumor segmentation. The network combined attention mechanisms, inception modules, and residual blocks with dilated convolutions to capture hierarchical features and spatial context efficiently. In BraTS2020, it achieved DSC of 87%, 84%, and 86% for WT, ET, and TC, respectively. Finally, Alanazi et al. [24] proposed M-BDCU-Net, which integrated bidirectional convolutional long- and short-term memory layers with densely connected convolutions. The network improved spatial feature learning and

captured temporal relationships across MRI slices to identify the ET region more precisely. It obtained a DSC of 80%, 82%, and 78% on the BraTS2020, and 81%, 85%, 82% on the BraTS2021 datasets for TC, ET, and WT, respectively.

Our proposed approach METAL addresses critical limitations in existing studies [13, 16, 17, 4, 25, 8, 18, 21, 19, 20, 22, 23, 24] by providing novel findings and effective solutions. It introduces a novel segmentation framework capable of accurately segmenting brain tumors even when one imaging modality is missing, which is rarely explored in prior literature. The architecture builds upon the nnUNet backbone and integrates a bi-level meta-learning strategy with attention-based fusion to enhance tumor segmentation. Specifically, the framework employs a bi-level training mechanism in which the inner loop adapts lightweight, modality-specific adapters to handle partial-modality inputs, while the outer loop updates the global network using complete multi-modality scans. This dual-loop design enables the model to effectively capture complementary features, ensuring robust segmentation under missing-modality conditions. To further strengthen performance, an adversarial modality discriminator is incorporated to encourage the encoder to learn modality-invariant latent representations. The inclusion of lightweight adapters allows rapid adaptation to available inputs, maintaining consistent and reliable segmentation performance.

3 Methodology

Our approach uses a shared-encoder nnUNet backbone with attention-based fusion to combine available modalities, lightweight adapter modules for fast adaptation to missing inputs, and an adversarial discriminator to promote modality-invariant representations. The training strategy follows a bi-level optimization scheme, where the inner loop adapts the adapters using partial-modality data, and the outer loop updates the global network using full-modality scans to ensure consistency and generalization. Figure 1 illustrates a high-level overview of our model. The following sections discuss the dataset, problem formulation, model architecture, adversarial regularization, and training procedure in detail.

3.1 Datasets

For this study, we utilized three benchmark datasets from the Brain Tumor Segmentation (BraTS) challenge series: BraTS2019, BraTS2020, and BraTS2021. These datasets provide 3D multi-parametric MRI (mpMRI) scans along with expert-annotated tumor segmentations in NIfTI format [26, 27, 28, 29].

BraTS2019. This dataset includes 626 glioma cases, encompassing both high-grade glioblastomas (HGG) and lower-grade gliomas (LGG). Each patient has four MRI sequences: T1, post-contrast T1 (T1ce), T2, and FLAIR. The scans were co-registered to the SRI24 template, resampled to a 1 mm³ voxel size, and skull-stripped to remove non-brain tissue.

Expert annotations are provided for three tumor sub-regions: ET, TC, and WT.

BraTS2020. The 2020 version contains 660 glioma cases and follows the same MRI modalities and preprocessing as BraTS2019. Tumor annotations again focus on ET, TC, and WT regions, allowing consistent longitudinal benchmarking across datasets.

BraTS2021. BraTS2021 expanded the cohort to 2040 patients and additionally includes a radiogenomic task to predict the MGMT promoter methylation status in glioblastoma. It retains the same MRI sequences and tumor sub-region annotations as previous years, while providing binary labels for the radiogenomic classification task.

It is worth noting that all three datasets include all four MRI modalities (i.e., T1, T1ce, T2, and FLAIR). However, to evaluate our model’s capability in sub-region (i.e., ET, TC, and WT) segmentation and classification tasks when handling missing modality inputs, we simulated missing modalities during both training and testing. Table 1 summarizes the key details of the datasets.

Table 1: Summary of BraTS benchmark datasets (2019, 2020, and 2021). The datasets consist of 3D multimodal MRI imaging.

Dataset	Train	Val	Test	#Patients
BraTS2019 [26, 27]	335	125	166	626
BraTS2020 [28]	369	125	166	660
BraTS2021 [29]	1251	219	570	2040

3.2 Problem formulation

In clinical practice, not all MRI modalities are always available for every patient due to scanner limitations, patient conditions, or time constraints [13]. We define the full set of modalities as $D_f = \{T1, T1ce, T2, FLAIR\}$ and the subset available for the n^{th} patient as $D_m \subseteq D_f$, where each modality m is represented as a volumetric tensor $x_n^{(m)} \in \mathbb{R}^{H \times W \times D}$ and the corresponding segmentation map is $Y_n \in \{0, 1, 2, 3\}^{H \times W \times D}$.

In this study, our goal is to learn a function f_θ parameterized by θ that maps any available subset $x_n^{(m)} : m \in D_m$ of X_m to an accurate segmentation prediction $\hat{Y}_n$ of $f_\theta(X_m)$, while ensuring the latent representations remain consistent across all combinations of available modalities. To achieve this, we adopt a meta-learning framework that separates rapid adaptation to missing modalities from global network optimization. Lightweight adapter parameters $\Delta\theta$ adjust the network features for partial-modality inputs X_m , enabling fast, sample-specific adaptation without modifying the global parameters θ , while a small set of full-modality scans X_f guides outer-loop optimization to maintain stable and generalizable latent representations. In essence, the inner loop

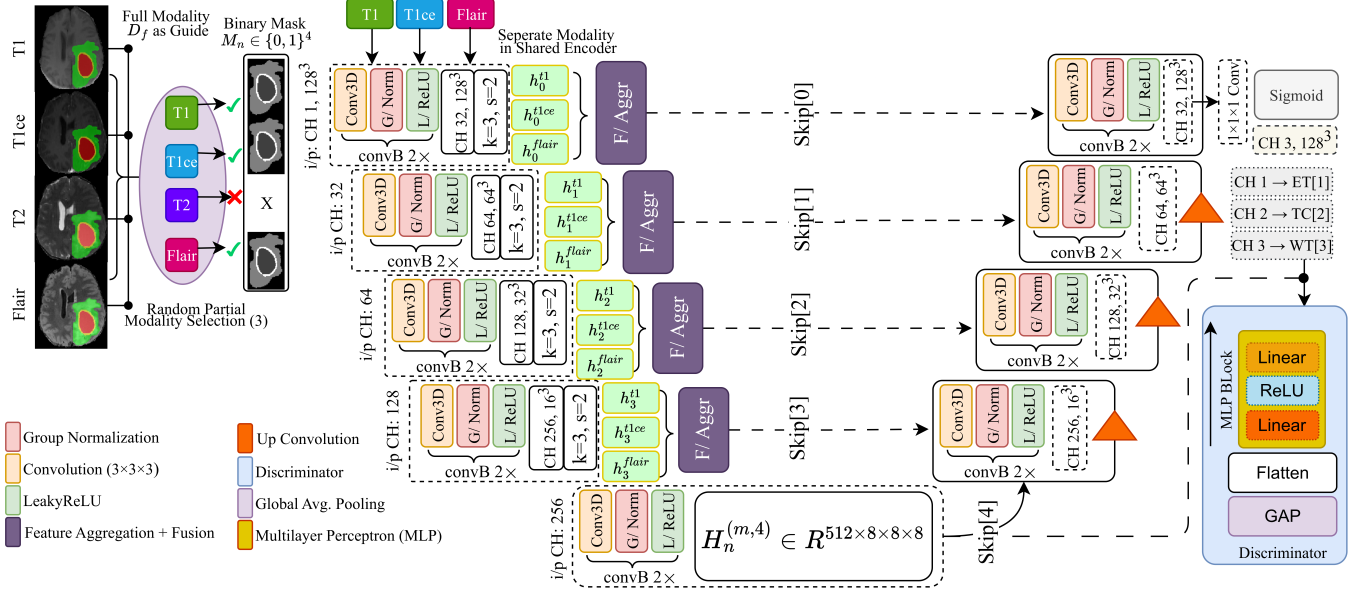


Figure 1: Overview of the proposed framework. The model processes three random available modalities (e.g., T1, T1ce, and FLAIR) independently through a shared encoder to generate modality-specific feature maps at multiple resolution levels. Missing modalities (e.g., T2) are handled via zero-padding and a binary modality mask. Features from available modalities are then fused using attention-based aggregation at each level. The aggregated features are decoded to produce a final segmentation map, while a discriminator receives the bottleneck features to enforce modality-invariant representations.

updates $\Delta\theta$ to handle incomplete modalities efficiently, and the outer loop refines θ to ensure consistency and robustness using full-modality data under variable MRI modality availability.

3.3 Model architecture

We built our segmentation framework upon a 3D nnUNet backbone [30], which efficiently handles multi-modal volumetric MRI data. For each modality $m \in D_f$, we process it independently through a dedicated encoder E_m to extract high-level, modality-specific features. Specifically, for the n^{th} patient, the volumetric input $x_n^{(m)}$ is passed through its corresponding encoder to produce a latent feature map h_m , as shown in Equation (1):

$$h_m = E_m(x_n^{(m)}), \quad h_m \in \mathbb{R}^{C \times H' \times W' \times D'} \quad (1)$$

where C denotes the number of feature channels and H', W', D' represent the reduced spatial dimensions after convolution and pooling. When a modality is missing for a patient, we replace $x_n^{(m)}$ with a zero tensor and use a binary mask M_m to indicate presence or absence. This mask guides both the meta-generator, which produces adaptive parameters for missing modalities, and the attention mechanism that we use to aggregate modality features.

After encoding each modality, we aggregate the information from the available modalities using attention-based feature weighting. We compute an attention weight w_m for each feature map h_m and then combine them to form the aggregated feature representation H , shown in Equation (2):

$$H = \sum_{m \in D_m} w_m \cdot h_m \quad (2)$$

This approach allows us to emphasize informative modalities and reduce the effect of missing or low-quality inputs, while producing concentrated feature representations regardless of which modalities are available.

We then feed the aggregated feature map H into the decoder to reconstruct the segmentation map at the original resolution. The output of the decoder S has dimensions corresponding to the number of classes, height H , width W , and depth D . The final segmentation, $\hat{Y} * n$, is obtained by taking the argmax over the class dimension of S . To handle partial-modality inputs efficiently, we incorporated lightweight adapter modules $\Delta\theta$ into both the encoder and decoder layers. These adapters are generated by a meta-generator $g * \phi$ conditioned on the input features and modality masks, defined as shown in Equation (3):

$$f_{\theta \oplus \Delta\theta}(X_m) = f_{\theta}(X_m) + g_{\phi}(X_m, M) \quad (3)$$

During the inner loop of meta-learning, we update only $\Delta\theta$, allowing rapid, sample-specific adaptation to missing

modalities while keeping the global network parameters θ fixed. This design also reduces computation and preserves stability.

3.4 Adversarial modality discriminator

We observed that the latent features produced by the encoder can become modality-specific [31]. This poses a challenge when certain modalities are missing during inference, as the network may fail to extract consistent semantic information [32].

To mitigate this, we introduced an adversarial modality discriminator, denoted as D_ψ and parameterized by ψ , which encourages the network to learn modality-invariant representations. The discriminator takes the aggregated feature map H (see Equation (2)) as input and predicts the presence or absence of each modality. For the n -th patient, we denote the binary true modality presence (i.e., 1 if modality m is available and 0 otherwise) as $y_{n,m}$. The predicted probability that m is present is given by Equation (4):

$$\hat{y}_{n,m} = D_\psi(H_n)_m \quad (4)$$

where $\hat{y}_{n,m} \in [0, 1]$ represents the discriminator's confidence for modality m in the n^{th} sample. To train the discriminator to correctly identify modality presence, we used a binary cross-entropy loss to encourage the discriminator to correctly identify which modalities are present in each input, as shown in Equation (5):

$$\begin{aligned} \mathcal{L}_{adv}^D = & -\frac{1}{N} \sum_{n=1}^N \sum_{m \in D_f} \left[y_{n,m} \log \hat{y}_{n,m} \right. \\ & \left. + (1 - y_{n,m}) \log(1 - \hat{y}_{n,m}) \right] \end{aligned} \quad (5)$$

where D_f is the set of all modalities. Simultaneously, we train the segmentation network along with the meta-generator and adapter modules to fool the discriminator and promote modality-invariant latent features. The adversarial loss from the generator perspective, $\mathcal{L}_{adv}^G$, is defined as the negative of the discriminator loss, $-\mathcal{L}_{adv}^D$, so that minimizing $\mathcal{L}_{adv}^G$ encourages the encoder and adapters to produce feature maps that obscure modality identity while retaining relevant tumor information. An overview of this adversarial feature regularization process is shown in Algorithm 1.

During the inner loop of meta-learning, we update only the adapter parameters $\Delta\theta$ to rapidly adapt to missing modalities. The update rule, as defined in Equation (6), combines the segmentation loss $\mathcal{L}_{seg}$ and the adversarial generator loss:

$$\Delta\theta \leftarrow \Delta\theta - \alpha \nabla_{\Delta\theta} \left\{ \mathcal{L}_{seg}(f_{\theta \oplus \Delta\theta}, X_m, Y_m) + \lambda_{adv} \mathcal{L}_{adv}^G(H) \right\} \quad (6)$$

where α is the inner-loop learning rate, X_m represents the partial-modality input, Y_m is the corresponding segmentation

Algorithm 1 Adversarial Regularization for Modality-Invariant Feature Learning

Input: Feature encoder E_m ; Meta-generator g_ϕ ; Base network f_θ ; Adapters $\Delta\theta$; Discriminator D_ψ ; Training data $\mathcal{D}$ w/ modalities D_f ; Loss weight λ_{adv} ; Learning rate α .

```

1: for each mini-batch  $(X, Y) \sim \mathcal{D}$  do
2:   for each patient  $n$  in batch do
3:     enumerate available subsets  $D_m \subseteq D_f$ 
4:     for each subset  $D_m$  do
5:       encode available modalities:  $h_m \leftarrow E_m(x_n^{(m)})$ 
6:       aggregate features:  $H \leftarrow \sum_{m \in D_m} w_m \cdot h_m$ 
7:       generate adapters:  $\Delta\theta_n \leftarrow g_\phi(H, M)$ 
8:       apply adapters  $\Delta\theta_n$  to  $f_\theta$ 
9:       forward pass  $\hat{Y}_n \leftarrow f_{\theta \oplus \Delta\theta_n}(X_m)$ 
10:      discriminator prediction:  $\hat{M} \leftarrow D_\psi(H)$ 
11:      segmentation loss  $\mathcal{L}_{seg}(\hat{Y}_n, Y_n)$ 
12:      discriminator loss  $\mathcal{L}_{adv}^D \leftarrow \text{BCE}(\hat{M}, M)$ 
13:      generator adversarial loss  $\mathcal{L}_{adv}^G \leftarrow -\mathcal{L}_{adv}^D$ 
14:      update adapters:  $\Delta\theta_n$ 
15:      update discriminator:  $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}_{adv}^D$ 
16:    end for
17:  end for
18: end for
```

Output: Modality-invariant feature representations H .

label, and λ_{adv} balances the segmentation and adversarial objectives. At the same time, the discriminator parameters ψ are updated to maximize $\mathcal{L}_{adv}^D$, ensuring it remains a strong adversary. In the outer loop, we update the global network parameters θ and meta-generator parameters ϕ using full-modality samples X_f , while continuing to enforce modality-invariant features and regularize the adapters. The outer loop update rule is defined in Equation (7):

$$\begin{aligned} \theta, \phi \leftarrow & \theta, \phi - \eta \nabla_{\theta, \phi} \left\{ \mathcal{L}_{seg}(f_{\theta \oplus \Delta\theta}(X_f), Y_f) \right. \\ & \left. + \lambda_{adv} \mathcal{L}_{adv}^G(H_f) + \lambda_{reg} \|\Delta\theta\|^2 \right\} \end{aligned} \quad (7)$$

where η is the outer-loop learning rate, Y_f is the ground-truth segmentation for full-modality inputs, and λ_{reg} controls the regularization on the adapters. Through this adversarial design, the network focuses on semantic tumor features while ignoring modality-specific cues.

3.5 Bi-level optimization and meta-learning updates

To handle missing modalities in the scans while maintaining high segmentation accuracy, we frame our learning problem as a bi-level optimization task. The key idea is to separate adaptation to partial modality inputs from the global network optimization, implemented through an inner loop and an

outer loop. An overview of the process is shown in Algorithm 2.

Algorithm 2 Bi-level Meta-Training for Modality Adaptation

Input: Partial-modality data $\mathcal{D}_m$; Full-modality data $\mathcal{D}_f$; Segmentation network f_θ w/ base params θ ; Lightweight adapter params $\Delta\theta$; Meta-generator g_ϕ ; Adversarial discriminator D_ψ ; Inner loop LR α , outer loop LR η ; Loss weights: λ_{adv} , λ_{reg} , total epochs E .

```

1: Initialize parameters:  $\theta, \Delta\theta, \phi, \psi$ 
2: for epoch  $\leftarrow 1$  to  $E$  do
3:   //Loop on the partial-modality data  $\mathcal{D}_m$ 
4:   for each mini-batch  $(X_m, Y_m) \sim \mathcal{D}_m$  do
5:      $\Delta\theta_0 \leftarrow \Delta\theta$  //Init adapters
6:     for  $k \leftarrow 1$  to  $K$  do //K-step inner adaptation
7:        $\hat{Y} \leftarrow f_{\theta \oplus \Delta\theta_{k-1}}(X_m)$ 
8:        $H \leftarrow \text{GetBottleneckFeatures}(X_m)$ 
9:       //Discriminator predicts modality presence
10:       $\hat{M} \leftarrow D_\psi(H)$ 
11:       $\mathcal{L}_{seg} \leftarrow \text{DiceLoss}(\hat{Y}, Y_m)$ 
12:      //Generator fools the discriminator
13:       $\mathcal{L}_{adv}^G \leftarrow -\mathcal{L}_{adv}^D = \text{BCE}(\hat{M}, \mathbf{1})$ 
14:       $\mathcal{L}_{inner} \leftarrow \mathcal{L}_{seg} + \lambda_{adv}\mathcal{L}_{adv}^G$ 
15:       $\Delta\theta_k \leftarrow \Delta\theta_{k-1} - \alpha \nabla_{\Delta\theta} \mathcal{L}_{inner}$ 
16:    end for
17:     $\Delta\theta^* \leftarrow \Delta\theta_K$ 
18:    //Adversarial component: update discriminator
19:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}_{adv}^D$ 
20:    Retrieve  $(X_f, Y_f) \sim \mathcal{D}_f$  //Outer Loop on  $\mathcal{D}_f$ 
21:     $\hat{Y} \leftarrow f_{\theta \oplus \Delta\theta^*}(X_f)$  //Forward pass with adapted net
22:     $H_f \leftarrow \text{GetBottleneckFeatures}(X_f)$ 
23:     $\hat{M}_f \leftarrow D_\psi(H_f)$ 
24:     $\mathcal{L}_{seg}^{outer} \leftarrow \text{DiceLoss}(\hat{Y}, Y_f)$ 
25:     $\mathcal{L}_{adv}^{G, outer} \leftarrow -\mathcal{L}_{adv}^D = \text{BCE}(\hat{M}_f, \mathbf{1})$ 
26:     $\mathcal{L}_{reg} \leftarrow \|\Delta\theta^*\|_2^2$  //L2 regularization on adapters
27:     $\mathcal{L}_{outer} \leftarrow \mathcal{L}_{seg}^{outer} + \lambda_{adv}\mathcal{L}_{adv}^{G, outer} + \lambda_{reg}\mathcal{L}_{reg}$ 
28:    //Update base + meta-generator
29:     $\theta, \phi \leftarrow \theta, \phi - \eta \nabla_{\theta, \phi} \mathcal{L}_{outer}$ 
30:  end for
31: end for

```

Output: Adaptive segmentation model f_{θ^*}

The inner loop updates only the lightweight adapter parameters $\Delta\theta$ for partial-modality inputs X_m , while keeping the global network parameters θ fixed. This allows the network to adjust its features to compensate for absent modalities without retraining the entire backbone. The update rule combines the segmentation loss $\mathcal{L}_{seg}$ and the adversarial generator loss $\mathcal{L}_{adv}^G$ (see Equation (6)) and encourages fast, sample-specific adaptation even when some modalities are unavailable. On the other hand, the outer loop (see Equation (7)) provides global guidance by updating the main network

parameters θ and the meta-generator parameters ϕ using full-modality samples X_f , while regularizing the adapter parameters and maintaining modality-invariant representations.

The process is visualized in Figure 2. The bi-level structure of METAL enforces stability and generalization: the outer loop anchors the network to complete modality information, while the inner loop allows flexible adaptation to incomplete inputs. The adversarial modality discriminator is integrated into both loops: during the inner loop, its loss informs adapter updates; and during the outer loop, it contributes to global parameter updates and ensures that latent features remain modality-invariant across both adaptation and global learning stages. Formally, the meta-learning procedure optimizes a combined objective as defined in Equation (8):

$$\mathcal{L}_G = \lambda_{seg}\mathcal{L}_{seg} + \lambda_{adv}\mathcal{L}_{adv}^G + \lambda_{reg}\|\Delta\theta\|^2 \quad (8)$$

where λ_{seg} weights the segmentation loss, λ_{adv} the adversarial generator loss, and λ_{reg} the regularization on the adapters. Each gradient step preserves the separation between fast-adapting components (adapters) and global components (base network and meta-generator). This design ensures stable convergence, enables generalization to unseen modality combinations, and provides the network with flexibility to handle missing modalities effectively.

3.6 Gradient derivation and analysis of meta-learning updates

To better understand the dynamics of our bi-level meta-learning framework, we derive the gradient updates for both the inner and outer loops [33]. Here, we denote the segmentation network with f_θ , which is parameterized by the base weights θ . The lightweight adapter modules are represented separately as $\Delta\theta$. For a partial-modality input X_m , the inner loop adapts the adapters by minimizing a task-specific objective that combines segmentation and adversarial losses, as defined in Equation (9):

$$\mathcal{L}_{inner}(\Delta\theta) = \mathcal{L}_{seg}(f_{\theta \oplus \Delta\theta}(X_m), Y_m) + \lambda_{adv}\mathcal{L}_{adv}^G(H). \quad (9)$$

where $\mathcal{L}_{seg}$ encourages voxel-wise agreement with the ground-truth segmentation mask Y_m , while $\mathcal{L}_{adv}^G$ promotes modality-invariant latent features H . The adapters are updated on $\Delta\theta$ by gradient descent, where the update subtracts α times the gradient of the inner loss $\mathcal{L}_{inner}$ to $\Delta\theta$. This way, the gradient signal comes jointly from the segmentation objective and the adversarial constraint and allows the adapters to specialize to balance for the absence of certain modalities.

After the inner-loop adaptation, the outer loop updates the global parameters θ and the meta-generator ϕ by leveraging full-modality samples X_f . The outer-loop objective is defined in Equation (10):

$$\mathcal{L}_{outer}(\theta, \phi) = \mathcal{L}_{seg}\{f_{\theta \oplus \Delta\theta^*}(X_f), Y_f\}$$

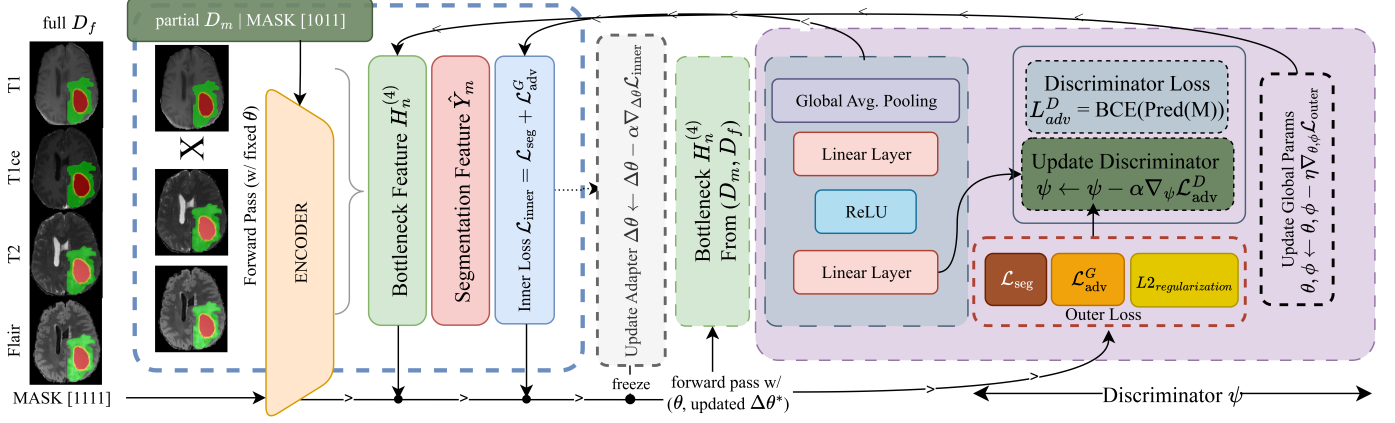


Figure 2: Bi-level meta-learning training strategy. The inner loop performs rapid adaptation on partial-modality data $\mathcal{D}_m$ by updating only the lightweight adapter parameters $\Delta\theta$. This enables fast, sample-specific adjustment to missing modalities while keeping the global network parameters θ fixed. The outer loop refines θ and the meta-generator ϕ using full-modality data $\mathcal{D}_f$. An adversarial modality discriminator ψ is updated in both loops to enforce modality-invariant features by predicting the presence of input modalities from bottleneck features.

$$+ \lambda_{adv} \mathcal{L}_{adv}^G(H_f) + \lambda_{reg} \|\Delta\theta^*\|^2 \quad (10)$$

where $\Delta\theta^*$ represents the adapted adapter parameters obtained from the inner loop. Unlike standard training, the gradient of this outer objective depends both directly on the base parameters and indirectly through their influence on $\Delta\theta^*$. Formally, the update rule for θ is given by Equation (11), while the update for ϕ is given by Equation (12):

$$\nabla_{\theta} \mathcal{L}_{outer} = \frac{\partial \mathcal{L}_{outer}}{\partial f} \frac{\partial f}{\partial \theta} + \frac{\partial \mathcal{L}_{outer}}{\partial f} \frac{\partial f}{\partial \Delta\theta^*} \frac{\partial \Delta\theta^*}{\partial \theta} \quad (11)$$

$$\nabla_{\phi} \mathcal{L}_{outer} = \frac{\partial \mathcal{L}_{outer}}{\partial f} \frac{\partial f}{\partial \Delta\theta^*} \frac{\partial \Delta\theta^*}{\partial \phi} \quad (12)$$

The first term in each update corresponds to the direct effect of the parameters on segmentation performance, whereas the second term accounts for the indirect effect that propagates through the inner-loop adaptation. This bi-level formulation ensures that the outer loop not only optimizes performance on complete inputs but also shapes the base parameters such that the network can quickly adapt to missing modalities.

Finally, the adversarial regularization plays an important role in enforcing modality invariance. The generator-side loss penalizes the discriminator D_{ψ} for correctly identifying which modalities are present from the latent features H . This constraint ensures that the segmentation relies on tumor-relevant structures that are shared across modalities, rather than on modality-dependent artifacts, thereby improving robustness in partial-modality scenarios.

4 Results and Experiments

4.1 Performance assessment

Segmentation performance achieved by the proposed model. As shown in Table 2, our proposed model, METAL, consistently produced reduced losses across both the outside and inner optimization loops. The outer generator, segmentation, and adversarial losses were 0.2503, 0.2175, and 0.1081, respectively, with the inner generator indicating a lower loss of 0.1944. The discriminator completed with a final loss of 0.1513, indicating that the generator and discriminator received balanced training throughout adversarial learning. This model demonstrated strong segmentation performance, achieving a mean DSC of 0.9561, a mean accuracy of 0.9973, and a mean F1-score of 0.9864.

Table 2: Quantitative performance metrics of the proposed segmentation model, including generator, discriminator, and evaluation losses, as well as accuracy-based measures.

Metric	Value
Outer Generator Loss	0.2503
Outer Segmentation Loss	0.2175
Outer Adversarial Loss	0.1081
Inner Generator Loss	0.1944
Discriminator Loss	0.1513
Mean Dice Coefficient	0.9561
Mean Accuracy	0.9973
Mean F1-score	0.9864

Training dynamics and hyperparameter. The METAL framework begins with a base feature dimensionality of 32, gradually increasing throughout the encoder to strengthen representational ability. In our training configurations, the intermediate representations are compressed using a bottleneck

scaling factor of 0.25, which reduces computational demand without discarding critical features. The training approach implemented a meta-learning strategy, having the inner loop perform a single gradient update for each task and gradient accumulation set to two steps to approximate higher batch sizes while following memory restrictions. Adversarial learning was regulated by giving the adversarial loss a weight of 0.1, resulting in a stable interaction between the generator and discriminator. The meta-generator and discriminator were both trained at a learning rate of 1.00E-04 for optimization, while the inner-loop adaptation used a higher rate of 1.00E-03 that allowed for rapid task-specific updates. The discriminator is designed with a hidden dimensionality of 48 to offer sufficient capacity to effectively distinguish between real and generated data. Table 3 details the hyperparameter configurations, and the hyperparameter selection process is further detailed in Section 4.4.

Table 3: Selected parameters for stable training dynamics and effective coordination between the generator and discriminator components. LR denotes the learning rate, λ represents the adversarial generator weighting coefficient, α refers to the inner-loop learning rate, and k indicates the number of inner-loop adaptation steps.

Category	Parameter	Value
Architecture	Base Features	32
	Bottleneck Scale	0.25
Training	Adversarial Generator (λ)	0.1
	Inner Loop Step (k)	1
	Accumulation Steps	2
Optimization	Meta Generator LR	0.0001
	Discriminator LR	0.0001
	Inner LR (α)	0.001
Adversarial	Discriminator Hidden Dim.	48

4.2 Experimental evaluation on the datasets

Segmentation results on BraTS2020 and BraTS2021.

The proposed framework is evaluated using BraTS2020 and BraTS2021 on three tumor subregions: WT, TC, and ET. The model achieves higher accuracy, precision, and recall on the BraTS2021 dataset compared to BraTS2020. Precision values are consistently high, exceeding 99% across all subregions, with an average of 99.32%. The recall values ranged from 97.64% to 98.14%, with an average of 97.87%. Similarly, the F1-scores varied between 98.47% and 98.77%, resulting in an overall average of 98.59%. Specificity remained consistently high, exceeding 99% across all subregions, with an overall average of 99.20%.

On the BraTS2021 dataset, the model demonstrated robust performance, achieving accuracies of 99.04% for WT, 98.77% for TC, and 97.94% for ET, resulting in an overall

average accuracy of 98.58%. Precision for all subregions remained between 97.81% and 98.74%, averaging 98.37%, while recall scores ranged from 97.90% to 98.36%, with an average of 98.13%. The F1-scores and specificity for BraTS2021 are higher, averaging 98.24% and 98.88%, respectively, across all subregions. These results, as reported in Table 4, highlight the robustness and reliability of the proposed model, METAL, demonstrating consistently strong segmentation performance for tumor subregions on both datasets.

DSC scores on BraTS2020 and BraTS2021. As presented in Table 5, each dataset comprises four imaging modalities: T1, T1ce, T2, and FLAIR, covering all tumor subregions. In BraTS2020, the WT subregion achieves scores ranging from 90.1% to 95.3% with an average of 92.6%, TC ranges from 88.5% to 94.8% with an average of 90.9%, and ET ranges from 85.1% to 91.6% with an average of 87.1%. Among the modalities, T1ce consistently achieves the highest performance, ranging from 91.6% to 95.3%, followed by T2 and FLAIR, while T1 records the lowest scores, ranging from 85.1% to 90.1%. In BraTS2021, all subregions show improvement: WT scores between 91.5% and 94.8% with an average of 93.3%, TC ranges from 89.3% to 93.9% with an average of 91.2%, and ET ranges from 86.9% to 91.0% with an average of 88.3%. Modality-wise, T1ce again performs best, ranging from 91.0% to 94.8%, while the remaining modalities range from 87.5% to 93.7%. These results demonstrate a consistent improvement in the data sets and robustness of the framework across sub-regions and imaging modalities. Figure 3 illustrates brain tumor segmentation, highlighting the tumor regions predicted by the model.

Performance on high and low-grade tumor subregions in BraTS2019. Further evaluation on BraTS2019 demonstrated the strong performance of the model in both HGG and LGG subregions. For high-grade cases, WT achieved 98.57% accuracy, 98.40% F1-score, 99.44% specificity, and a DSC of around 92.31%. TC reached 98.19% accuracy, 98.34% F1-score, 99.31% specificity, and a DSC of around 90.53%, while ET showed slightly lower performance with 98.03% accuracy, 97.88% F1-score, 99.16% specificity, and a DSC of around 88.37%. Averaged across all subregions, high-grade cases attained 98.26% accuracy, 98.21% F1-score, 99.30% specificity, and a DSC of around 90.40%.

Low-grade cases demonstrated consistent performance, with WT achieving 97.94% accuracy, 97.90% F1-score, 98.87% specificity, and a DSC of approximately 90.62. TC showed slightly lower performance, with 97.82% accuracy, 97.41% F1-score, 97.74% specificity, and a DSC reaching 88.71%, while ET performed the lowest among the subregions, reaching 97.01% accuracy, 96.67% F1-score, 97.93% specificity, and a DSC of roughly 85.92. Overall, low-grade cases attained 97.59% accuracy, 97.73% F1-score, 98.18% specificity, and a DSC of around 88.42. From Table 6, it can be stated that low-grade tumor images performed lower

Table 4: Subregion classification metrics for BraTS2020 and BraTS2021. For each subregion, the average results obtained across the four modalities are reported.

Dataset	Subregion	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)
BraTS2020	WT	98.30	99.24	97.82	98.52	99.53
	TC	98.66	99.40	98.14	98.77	99.11
	ET	98.72	99.31	97.64	98.47	98.97
	Avg.	98.56	99.32	97.87	98.59	99.20
BraTS2021	WT	99.04	98.74	98.36	98.55	99.18
	TC	98.77	98.55	98.12	98.33	98.92
	ET	97.94	97.81	97.90	97.85	98.53
	Avg.	98.58	98.37	98.13	98.24	98.88

Table 5: Modality- and subregion-wise DSC scores of the proposed model for BraTS2020 and BraTS2021 datasets. The best and second-best results for each dataset are **bolded** and underlined, respectively.

Dataset	Subregion	Modalities				Avg
		T1	T1ce	T2	Flair	
BraTS2020	WT	90.12±1.23	95.34±0.98	92.28±1.15	92.62±1.05	92.59±1.10
	TC	88.45±1.47	94.76±0.85	90.88±1.59	89.35±1.31	90.86±1.31
	ET	85.24±1.75	91.62±1.20	86.51±1.51	<u>85.12±1.40</u>	87.12±1.46
	Avg	87.94±1.48	93.91±1.01	89.89±1.42	<u>89.03±1.25</u>	-
BraTS2021	WT	91.50±1.12	94.80±0.95	93.25±1.05	93.70±1.00	93.31±1.03
	TC	89.30±1.33	93.85±1.11	91.15±1.25	90.50±1.19	91.20±1.23
	ET	87.50±1.40	91.00±1.31	87.75±1.49	<u>86.90±1.35</u>	88.29±1.39
	Avg	89.43±1.28	93.22±1.12	90.72±1.26	<u>90.37±1.18</u>	-

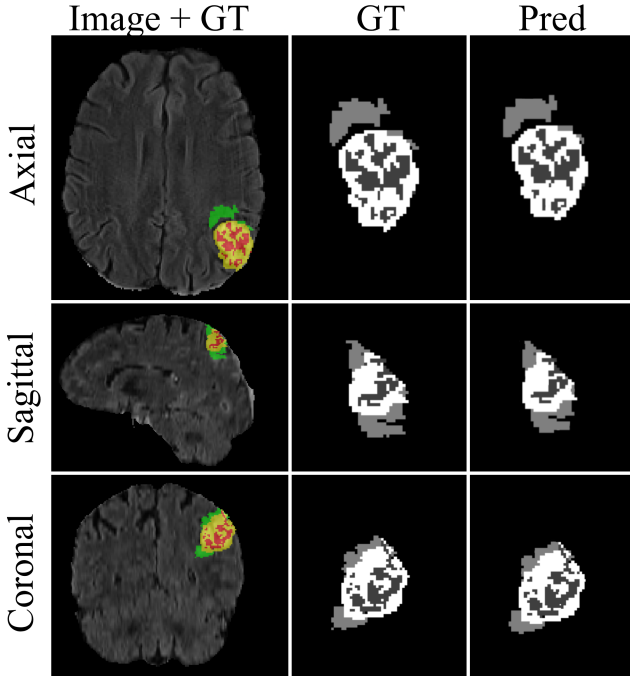


Figure 3: Comparison of ground truth (GT) and predicted tumor segmentation on the BraTS2020 dataset.

DSCs compared to high-grade tumor images. Figure 4 illustrates the appearance of brain tumors in four different MRI modalities (T1, T1ce, T2, and FLAIR).

Table 6: Performance comparison of high-grade and low-grade tumor subregions of BraTS2019 dataset.

Grade	Sr.	Accuracy	F1-score	Specificity	DSC
High	WT	98.57	98.40	99.44	92.31 ± 0.17
	TC	98.19	98.34	<u>99.31</u>	90.53 ± 1.38
	ET	98.03	97.88	99.16	88.37 ± 3.72
	Avg.	98.26	98.21	99.30	90.40 ± 1.76
Low	WT	97.94	97.90	98.87	90.62 ± 1.43
	TC	97.82	97.41	97.74	88.71 ± 0.93
	ET	97.01	96.67	97.93	85.92 ± 3.94
	Avg.	97.59	97.33	98.18	88.42 ± 2.10

4.3 Performance comparison

4.3.1 Comparison with existing methods

We compared the performance of our model on several datasets, including BraTS2019 (HGG), BraTS2019 (LGG), BraTS2020, and BraTS2021, with R2U++ [34], MUNet [35], U-Net++ [36], Attention U-Net [37], nnUNet (without meta-learning) [30], UNETR [38], TransUNet [39], and Swin-Unet [40], as shown in Table 7. In most cases, our method outperformed the other approaches. For instance, on BraTS2019

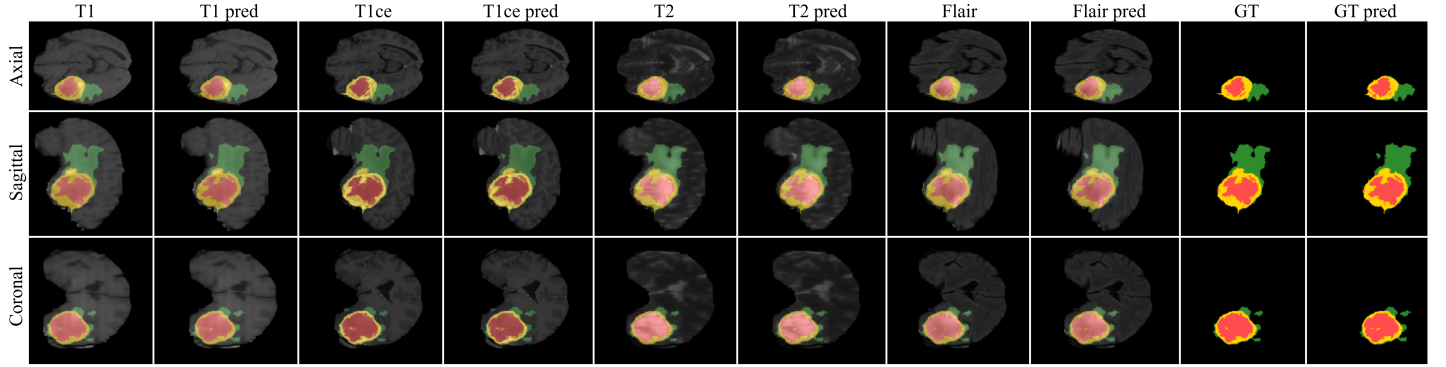


Figure 4: Visualization of brain tumor regions across the four MRI modalities (T1, T1ce, T2, and FLAIR) from the BraTS2019 dataset.

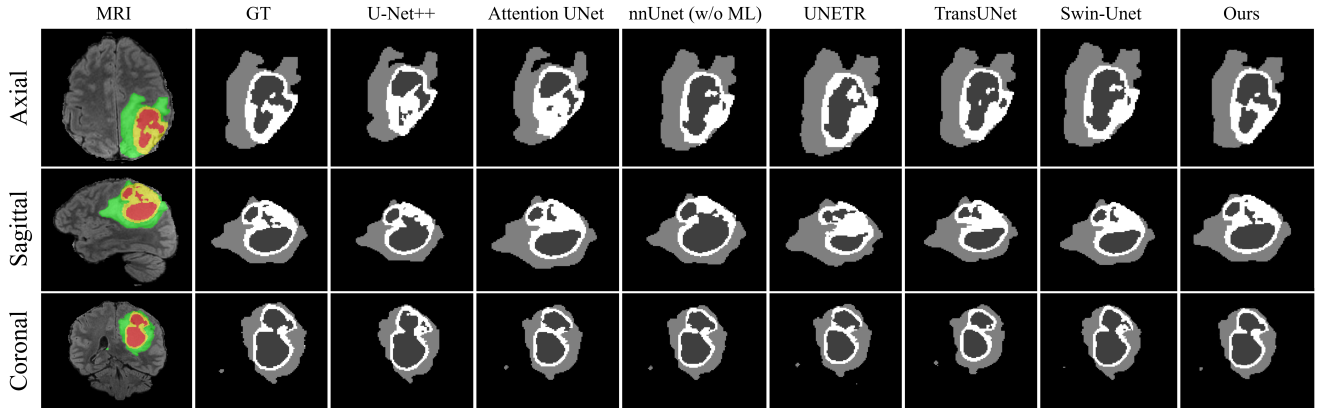


Figure 5: Comparative performance of the proposed METAL and baseline models, including UNet++, Attention U-Net, nnUNet (without meta-learning), UNETR, TransUNet, and Swin-Unet for brain tumor segmentation on the BraTS2021 dataset.

(HGG), it achieved around 92.31% for WT, approximately 90.53% for TC, and about 88.37% for ET, surpassing all other methods. On BraTS2019 (LGG), the model also maintained strong performance, achieving about 90.62% for WT, around 88.71% for TC, and approximately 85.92% for ET, although Swin-Unet [40] (approx. 91.82%) and nnUNet (without meta-learning) [30] (approx. 89.37%) performed slightly better in WT and TC, respectively. In contrast, on BraTS2020, it outperformed the other methods, obtaining around 92.59% for WT, approximately 90.86% for TC, and about 87.12% for ET. On BraTS2021, it reached about 93.31% for WT, around 91.20% for TC, and approximately 88.29% for ET, with slightly lower performance in TC compared to Swin-Unet [40], which achieved about 91.76%.

Overall, the model consistently attained the highest or nearly the second-highest scores across all sub-regions, demonstrating accurate brain tumor segmentation. Figure 5 illustrates that our method achieves more accurate and reliable segmentation results than the SOTA approaches.

4.3.2 Comparison with SOTA architecture

We compare the computational complexity and architectural specifications of our model with several existing approaches, including nnUNet [30], SwinUNETR [40], UNETR [38], TransUNet [39], MedSegDiff-V2 [41], MedSegDiff [42], and SegDiff [43]. Standard models such as nnUNet [30] use 16M parameters with 42.7G FLOPs, relying solely on CNN-based feature extraction. In comparison, transformer-based architectures such as SwinUNETR [40] and UNETR [38] require substantially greater computational resources, with 138M and 104M parameters and 312.4G and 278.9G FLOPs, respectively. TransUNet [39] uses 96M parameters and 245.6G FLOPs using a hybrid CNN-Transformer feature extractor. On the other hand, diffusion-based models, including MedSegDiff-V2 [41], MedSegDiff [42], and SegDiff [43], range from 23M to 46M parameters and 82.1G to 118.3G FLOPs, employing U-Net architectures combined with diffusion mechanisms as feature extractors.

Table 7: Comparative performance (in %) of our model and SOTA methods on the BraTS2019 (high- and low-grade), BraTS2020, and BraTS2021 datasets for brain tumor segmentation. The best and the second-best results for each subregion are bolded and underlined, respectively.

Dataset	Sub.	Ours	R2U++ [34]	MUNet [35]	U-Net++ [36]	AttUNet [37]	nnUNet [30]	UNETR [38]	TransUNet [39]	Swin-U [40]
BraTS19(HGG)	WT	92.31±0.17	88.79±0.25	87.45±0.28	87.71±0.34	88.92±0.27	89.20±0.24	90.07±0.29	89.15±0.26	87.10±0.31
	TC	90.53±1.38	86.82±1.42	85.65±1.48	86.91±1.05	87.14±1.45	87.55±1.41	85.09±1.50	87.40±1.43	85.45±1.52
	ET	88.37±3.72	84.05±3.80	83.12±3.88	83.85±3.85	85.66±3.75	85.65±3.70	84.95±3.78	85.20±3.72	83.60±3.90
BraTS19(LGG)	WT	90.62±1.43	86.25±1.59	85.05±0.55	85.40±1.55	86.08±1.52	87.10±1.48	85.20±1.58	86.95±1.19	91.81±1.62
	TC	88.71±0.93	84.50±0.98	83.35±1.05	83.90±1.08	85.44±1.02	89.37±0.99	83.80±1.17	85.73±0.11	87.04±1.12
	ET	85.92±3.94	81.35±2.08	80.25±2.05	81.20±4.08	82.10±4.01	82.55±3.95	81.97±3.11	<u>84.09±3.97</u>	83.85±3.90
BraTS20	WT	92.59±1.10	88.75±1.15	87.74±1.25	87.90±1.25	88.95±0.29	89.25±1.18	87.64±0.22	<u>89.08±1.15</u>	87.05±0.43
	TC	90.86±1.31	86.97±1.35	85.91±1.45	86.55±1.45	87.35±1.41	87.80±1.38	85.95±1.42	<u>89.65±1.33</u>	85.50±2.51
	ET	87.12±1.46	83.71±2.07	82.35±0.67	83.25±2.67	84.27±0.55	<u>84.93±2.52</u>	84.15±1.55	<u>84.45±1.50</u>	82.90±1.62
BraTS21	WT	93.31±1.03	89.27±1.05	87.95±1.17	88.40±1.22	89.57±1.08	89.04±1.05	88.05±2.08	90.17±1.05	92.75±1.15
	TC	91.20±1.23	87.25±1.25	85.95±3.34	86.75±0.31	87.85±1.28	88.25±1.25	86.60±1.28	88.35±1.25	91.76±2.35
	ET	88.29±1.39	84.76±1.44	82.95±1.45	83.75±1.45	84.85±1.42	85.25±1.40	84.55±1.42	85.00±1.38	<u>86.30±1.45</u>

Table 8: Comparative architectural overview of model architecture, including individual components parameters, FLOPs, input dimensions, feature extractors, and adapter configurations.

Category	Model	Params (M)	FLOPs (G)	Input Shape	Feature Extractor	Adapter
Standard	nnUNet [30]	16	42.7	4 × 128 × 128 × 128	CNN	×
	SwinUNETR [40]	138	312.4	4 × 128 × 128 × 128	Swin Transformer	×
	UNETR [38]	104	278.9	4 × 128 × 128 × 128	Vision Transformer	×
	TransUNet [39]	96	245.6	4 × 128 × 128 × 128	Hybrid CNN-Transformer	×
	MedSegDiff-V2 [41]	46	118.3	4 × 128 × 128 × 128	U-Net + Diffusion	×
	MedSegDiff [42]	25	87.6	4 × 128 × 128 × 128	U-Net + Diffusion	×
	SegDiff [43]	23	82.1	4 × 128 × 128 × 128	U-Net + Diffusion	×
METAL (ours)	Generator	34	95.2	4 × 128 × 128 × 128	Cross-Modal Attn	0.25 compression
	Modality Discriminator	2.1	1.8	512 × 8 × 8 × 8	3-layer MLP	-
	Adapter Block (per block)	0.12	0.04	Variable	Bottleneck FFN	0.75× reduction
	Final Model	51.3	97.5	4 × 128 × 128 × 128	Multi-Modal Fusion	Hierarchical 0.25-0.75×

In contrast, our model achieves a balance between performance and computational efficiency. The generator comprises 34M parameters and 95.2G FLOPs, incorporating cross-modal attention with a 25% compression factor, along with a modality discriminator with 2.1M parameters, 1.8G FLOPs, and a 3-layer MLP. Each adapter block contributes 0.12M parameters with 0.04G FLOPs and applies a 0.75× reduction in the bottleneck feedforward networks. The final model comprises 51.3M parameters and 97.5G FLOPs, integrating multi-modal features through hierarchical adapters with compression ranging from 0.25× to 0.75×. Overall, the model demonstrates both computational efficiency and effective multi-modal feature integration, while requiring substantially fewer resources than transformer-heavy architectures, as presented in Table 8.

4.3.3 Comparison with existing literature

Our proposed method METAL framework, outperforms existing models by achieving the highest DSC, as seen in Table 9. Previous models, such as MLA [13], achieved DSC scores over 80% for the WT and TC; however, their performance dropped below 70% for the ET. In contrast, ROI-DA [20] maintained consistent results above 85 across all three tumor regions. HIOFN [25] showed a fluctuation ranging from 63% to 84% for the three tumor regions, while GAIR-U-NET [23]

Table 9: Comparison of subregion DSC scores (WT, TC, ET) on BraTS datasets for methods reported in existing literature. The best and the second-best results in each subregion are bolded and underlined, respectively, for the corresponding datasets. All values are provided in percentage (%).

Ref.	Dataset	Model	Subregion DSC Score		
			WT	TC	ET
[13]	BraTS20	MLA	88.74	81.63	65.27
	BraTS18		87.12	79.12	62.53
[20]	BraTS20	ROI-DA	89.13	88.97	87.47
	BraTS20		84.97	74.42	63.83
[8]	BraTS20	LS3M	88.22	79.76	63.60
	BraTS18		91.38	88.27	82.44
[21]	BraTS19	MPEDA-Net	<u>91.87</u>	<u>88.71</u>	83.73
	BraTS20		<u>91.00</u>	<u>83.00</u>	81.00
[19]	BraTS21	MS-SegNet	<u>92.00</u>	84.00	<u>86.00</u>
	BraTS20		87.00	84.00	<u>86.00</u>
[23]	BraTS20	GAIR-U-Net	78.00	80.00	82.00
	BraTS21		82.00	81.00	85.00
Ours	BraTS19 (HGG)	METAL	92.31	90.53	88.37
	BraTS19 (LGG)		90.62	<u>88.71</u>	85.92
	BraTS20		92.59	90.86	87.12
	BraTS21		93.31	91.20	88.29

achieved 87%, 84%, and 86% for WT, TC, and ET, respectively. MPEDA-Net [21], MS-SegNet [19], and M-BDCU-Net [24] were evaluated on two versions of the BraTS dataset, yet none were able to surpass the performance of our method.

4.4 Ablation study

We conducted a systematic ablation study to evaluate the effects of input modalities, feature dimensions, input sizes, learning rates, adversarial loss weights, and hidden dimensions on segmentation performance, aiming to identify the combination that achieved the optimal average DSC, as presented in Table 10.

Different combinations of T1, T1ce, T2, and FLAIR sequences are evaluated to determine the contribution of each modality. Feature dimensions f_{dim} ranged from 16 to 64, and input volumes varied between 64^3 and 256^3 . The meta-generator learning rate is set between 0.0001 and 0.0005, while the inner loop learning rate varies from 0.001 to 0.005. Adversarial generator loss weights ($\lambda_{adv.g}$) are adjusted between 0.05 and 0.5, with hidden dimensions ranging from 32 to 64.

Among all configurations, the combination that excludes T1, however, included T1ce, T2, and FLAIR, with a feature dimension of 64 and an input size of 256^3 , achieves the highest DSC of 86.2%. However, this configuration involves a relatively high feature dimension. In contrast, using the same modalities with different feature dimensions, input sizes, and hidden dimensions results in reduced performance, with DSC values decreasing to 74.8%, indicating that small models with low-resolution inputs lose boundary precision. Therefore, we select configuration 8 for our model training, with a feature dimension of 32 and an input size of 128^3 , achieving a DSC of 84.5%. This configuration provides the best trade-off between segmentation performance and computational efficiency. These findings highlight the critical role of sufficient feature representation and demonstrate that carefully tuned hyperparameters are essential for achieving optimal segmentation accuracy.

5 Discussion

Our proposed model comprises several advanced components designed to enhance both its robustness and effectiveness. This incorporates a bi-level optimized meta-learning framework that integrates the nnUNet backbone, attention-based fusion, an adversarial modality discriminator, and lightweight adapters to efficiently handle missing MRI modalities, a challenge frequently encountered in real-world clinical environments for brain tumor segmentation. The adversarial modality discriminator further guides the network to learn modality-invariant latent representations, encouraging the encoder to focus on tumor-relevant features rather than modality-specific patterns. To enable rapid, sample-specific adaptation without the need to retrain the entire network, lightweight adapters are incorporated. A key strength of the model lies in its bi-level training strategy, the inner loop fine-tunes the adapters for partial-modality inputs. In contrast, the outer loop updates the global network using full-modality scans,

ensuring robust generalization across diverse combinations of available MRI modalities.

We assess our model on three datasets, including BraTS2019, BraTS2020, and BraTS2021. On BraTS2020, WT achieves a DSC of 92.59%, with T1ce providing the highest performance among the four modalities, followed by FLAIR, T2, and T1. For ET and TC, the average DSCs are 90.86% and 87.12%, respectively, with T1ce again delivering the best results in both cases. On BraTS2021, the performance further improves, with DSCs of 93.31%, 91.20%, and 88.29% for WT, TC, and ET, respectively. Consistent with BraTS2020, T1ce achieves the highest DSC across all three subregions, while the lowest values are observed in T1 for WT and TC, and in FLAIR for ET.

The effectiveness and robustness of the proposed model are further demonstrated on the BraTS2019 dataset. For BraTS2019 (HGG), the DSCs for WT, TC, and ET are 92.31%, 90.53%, and 88.37%, respectively, with an average of 90.42%. In contrast, for BraTS2019 (LGG), the DSCs slightly decrease to 90.62%, 88.71%, and 85.92% for WT, TC, and ET, respectively, resulting in an average of 88.42%. Among all datasets, BraTS2021 achieves the highest DSCs for WT, TC, and ET, whereas BraTS2019 (LGG) records the lowest values across the three subregions.

The proposed model comprises a generator, modality discriminator, and lightweight adapter blocks, with a total of 51.3 million parameters and 97.5% memory efficiency. It utilizes cross-modal attention and multi-modal fusion, with hierarchical compression strategies to reduce computational cost while preserving high segmentation performance. We conduct a series of experiments to determine the configuration that achieves optimal performance while maintaining moderate computational cost, as shown in Table 10. In earlier experiments, we tested a setup excluding T1 but including T1ce, T2, and FLAIR, with a feature dimension of 64 and an input size of 256^3 , which achieved a maximum DSC of 86.2%. However, this configuration requires relatively high computational resources. Our optimal configuration retains the same set of modalities but reduces the feature dimension to 32 and the input size to 128^3 , with meta-generator learning rates of 0.0001 and 0.0005, and an inner-loop learning rate of 0.001, achieving a DSC of 84.5% while significantly lowering computational demand.

Compared to recent SOTA models, METAL consistently demonstrates superior segmentation performance, achieving the highest DSC across all subregions in the three benchmark datasets. Models such as R2U++ [34], MUNet [35], and UNet++ [36] struggle to surpass the 90% threshold, while SwinUNet [40] achieves approximately 91.81%. Within existing studies, some methods, such as MPEDA-Net [21] and MS-SegNet [19], report DSC values above 90% for WT; however, others, including HIOFN [25] and LS3M [8], perform considerably lower, in some cases failing to reach even 65%. This

Table 10: Ablation study results showing the effects of input modalities, feature dimensions (f_{dim}), input sizes, learning rates, adversarial loss weights ($\lambda_{\text{adv-g}}$), and hidden dimensions (Hid Dem) on segmentation performance. The optimal configuration in terms of the best trade-off is highlighted.

Exp	T1	T1ce	T2	FLAIR	f_{dim}	Input	LR (Meta)	LR (Inner)	$\lambda_{\text{adv-g}}$	Hid Dem	DSC (%)
1	✗	✓	✓	✓	64	256 ³	0.0001	0.001	0.1	64	86.2
2	✓	✗	✓	✓	64	192 ³	0.0003	0.001	0.08	48	80.4
3	✓	✓	✗	✓	32	128 ³	0.0005	0.003	0.1	64	78.3
4	✗	✓	✓	✗	32	64 ³	0.0005	0.001	0.1	64	77.2
5	✓	✓	✓	✗	32	192 ³	0.0001	0.005	0.05	64	81.3
6	✓	✗	✓	✓	32	128 ³	0.0005	0.005	0.1	32	77.6
7	✗	✓	✓	✓	16	64 ³	0.0003	0.001	0.1	32	74.8
8	✗	✓	✓	✓	32	128 ³	0.0001	0.001	0.1	48	84.5
9	✓	✓	✗	✓	64	256 ³	0.0001	0.003	0.5	32	83.1
10	✗	✓	✓	✓	32	128 ³	0.0003	0.003	0.05	48	76.4
11	✓	✗	✓	✓	32	64 ³	0.0001	0.001	0.1	48	79.7

comparison clearly demonstrates the reliability and robustness of the proposed model.

Although METAL achieves promising performance, it presents several limitations. First, the model is specifically trained on MRI data, and its performance may decline when applied to other imaging modalities such as computed tomography (CT) or ultrasound scans. Second, although the model is robust to some missing modalities, its performance may deteriorate when multiple critical modalities are absent simultaneously. Third, the model may perform less accurately on tumor types or sub-regions that are underrepresented in the training datasets. Finally, even with optimizations such as lightweight adapters and hierarchical compression, METAL demands considerable computational resources, which could restrict its applicability in settings with limited hardware availability.

6 Conclusion

We present METAL, a highly effective framework for brain tumor segmentation from MRI, achieving consistently high DSC on three datasets. For WT, TC, and ET, METAL achieves 92.31%, 90.53%, and 88.37% for HGG, and 90.62%, 88.71%, and 85.92% for LGG on BraTS2019. On BraTS2020, the corresponding scores are 92.59%, 90.86%, and 87.12%, and on BraTS2021, performance further improves to 93.31%, 91.20%, and 88.29%. These results show that METAL has a strong generalization and superior ability to segment tumors across different datasets and tumor types. Furthermore, it remains effective even when certain modalities are unavailable, a situation frequently encountered in clinical practice.

The significance of this work lies in the integration of several advanced modules that collectively enhance the performance and robustness of METAL. The framework strategically combines lightweight adapters that adapt to available

input features, an adversarial modality discriminator that enforces modality-independent representations, and attention-based fusion with the nnUNet backbone within a bi-level meta-learning framework. In this bi-level training strategy, the inner loop refines the adapters for partial-modality inputs, while the outer loop updates the global network with full-modality scans. This integration of architectural innovations and training strategies allows METAL to consistently deliver high segmentation performance.

While METAL demonstrates strong performance, further work is needed to improve generalization to other modalities such as CT and ultrasound, underrepresented tumor types, and in different clinical domains. Improvements in computational efficiency are also required for practical clinical deployment. By providing accurate segmentation even when a modality is absent, METAL can support radiologists in diagnosis and treatment planning, contributing to better patient management.

Declarations

Conflict of Interests. The authors declare that they have no financial conflicts of interest that could have influenced this work.

Ethics Approval and Consent to Participate. No additional ethics approval or consent was required.

Dataset Availability. All datasets used in this study are publicly available.

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